

Md Shahanur Islam Shagor

Full-Stack Developer & AI Engineer

Detail-oriented and innovative **Software Engineer** with a specialized focus on **Autonomous Systems, UVA coordination**, and **Cybersecurity**. Proven experience in developing secure, scalable web applications and research-driven projects involving **Blockchain frameworks and AI-based computer vision**. Adept at implementing **post-quantum cryptography and decentralized protocols** to solve complex technical challenges. A strong problem-solver committed to continuous learning and advancing the field of autonomous technology.

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EDUCATION

Bachelor of Science

Voronezh State University of Forestry and Technology, Russia

Computer Science and Microelectronics Engineering

2023-Continue

Diploma in Engineering

Dhaka Politechnic Institute, Bangladesh.

Computer Technology

2016-2020

SKILLS

Backend

- PHP
- Larevel
- Node.js
- Python
- Flask
- Rest APIs
- GraphQL
- Web Socket
- GO
- C++
- C#

WORK EXPERIENCE

Web Developer (Remote)

Explo-IT | Bangladesh 2023 - 2025

- Designed and developed scalable web applications using Laravel (backend) and Next.js (frontend).
- Built and integrated RESTful APIs for seamless data exchange between frontend and backend systems.
- Implemented authentication, role-based access control, and secure payment gateways in Laravel.
- Optimised application performance via Eloquent query tuning, caching strategies, and database indexing.
- Created responsive, SEO-friendly UIs with Next.js, Tailwind CSS, and modern JavaScript frameworks.
- Delivered real-time features (chat, notifications, live updates) using WebSockets / Pusher.
- Deployed and maintained applications on Linux servers (Nginx, Apache, Docker) with CI/CD pipelines.
- Integrated third-party services: payment gateways, social logins, and cloud storage.

Web Developer (Onsite)

DataSoft LTD | Bangladesh 2021 - 2022

- Assisted in developing web applications using Laravel and Next.js.
- Built and maintained APIs, database queries, and responsive UI components.
- Collaborated with senior developers on bug fixing, code reviews, and feature enhancements.
- Implemented frontend layouts with Tailwind CSS and supported database migrations in Laravel.
- Wrote clear documentation and contributed to team knowledge-sharing sessions.

PROJECTS

1. WVAB: Wireless Vision-Aid System for the Blind
   - Technologies: Python, YOLOv8n, ROS, NVIDIA Jetson Nano, ESP32-CAM, UDP Streaming.
   - Developed an assistive navigation system utilizing YOLOv8n for real-time object detection with low-latency multilingual audio feedback.
   - Optimized UDP video streaming from ESP32-CAM to Jetson Nano, achieving high frame rates for obstacle recognition in a Linux/ROS environment.

   Implemented a custom spatial awareness algorithm to guide users through complex indoor environments.

Frontend

- React,js
- Next.js
- Vue.js
- Angular.js
- React Native
- TypeScript
- JavaScript
- Tailwind CSS
- Bootstrap
- HTML & CSS

Database

- SQLite
- Mysql
- PostgreSQL
- MongoDB
- Redis

DevOps & Tools

- Linux
- Nginx
- Apache
- Docker
- CI/CD Pipelines
- GitHub
- CMake

Mobile & Embedded

- Flutter
- Android Studio
- Raspberry Pi
- NVIDIA Jetson Nano
- ESP32-CAM
- Linux/ROS
- LiDAR

AI & Security

- YOLOv8n
- Blockchat
- Zero-Knowledge Proofs
- OWASP
- JWT / OAuth 2.0
- Post-Quantum Cryptography
- Cyber Security

Cloud & Infrastructure

- AWS (EC2, S3, Lambda)
- Google Cloud
- Digital Ocean
- Kubernetes
- Prometheus
- Grafana

Testing & QA

- Jest
- PHPUnit
- PyTest

2. Privacy-Preserving Blockchain for Autonomous Vehicles
   - Technologies: Go/Python, Zero-Knowledge Proofs (ZKP), SHA3-256, Post-Quantum Cryptography.
   - Designed a decentralized security framework for V2X (Vehicle-to-Everything) communication, focusing on identity privacy and data integrity.
   - Integrated Pedersen Commitments (ZKP) to validate vehicle transactions without revealing sensitive location or identity data.

   Researched and implemented Post-Quantum Cryptographic primitives to ensure long-term resilience against quantum-level threats.
3. SyncChat – Real-Time Communication Engine
   - Technologies: Node.js, React Native, WebSocket, WebRTC, Redis (Pub/Sub).
   - Built a high-concurrency chat engine supporting real-time messaging, live user presence, and peer-to-peer video calling via WebRTC.
   - Leveraged Redis Pub/Sub to handle message broadcasting and state management across multiple server instances.

   Developed a cross-platform mobile interface using React Native, ensuring sub-100ms message delivery latency.
4. Multivendor E-commerce Management Platform
   - Technologies: Laravel, Next.js, GraphQL, MySQL, Stripe, Nginx.
   - Engineered a scalable marketplace architecture supporting multiple vendor tiers, dynamic inventory management, and role-based access control (RBAC).
   - Integrated GraphQL for optimized data fetching, reducing frontend load times by 40%.

   Automated deployment using CI/CD pipelines and configured Nginx for load balancing and SSL termination.
5. Server Monitor — Real-Time Health Dashboard
   - Technologies: React, TypeScript, Node.js, Express, Socket.IO, Python Flask, psutil, SQLite, Docker, Nginx.
   - Developed a TypeScript + Socket.IO backend with a Python Flask agent for live CPU, RAM, and disk monitoring using psutil.
   - Implemented role-based authentication, SQLite persistence, Docker deployment, and automated health-check scheduling.

   Designed a responsive dashboard with live metrics, alert triggers, heartbeat monitoring, and failover status visualization.
6. Decentralized Coordination in Secure Autonomous Drone Swarms
   - Technologies: Python, C++, Byzantine Resilience, SHA3-256, Mesh Networking.
   - Developed a decentralized leader election and coordination framework for drone swarms, ensuring operational continuity despite node failures.
   - Implemented Byzantine-resilient algorithms and secure communication protocols using SHA3-256 chaining to prevent malicious takeovers within the swarm.

   Optimized high-concurrency swarm management using Go, focusing on secure data exchange over decentralized mesh networks.
7. Acoustic Localization for GPS-Denied Environments
   - Technologies: Python, C++, Go, CMake, Byzantine Resilience, SHA3-256, AES-256, Mesh Networking.
   - Engineered a robust localization system for autonomous UAVs operating in GPS-denied environments using acoustic signal processing.
   - Implemented TDOA (Time Difference of Arrival) algorithms to calculate precise coordinates based on sound wave propagation.

   Utilized Go for analyzing correlation matrices and noise filtering to improve localization accuracy in high-interference outdoor settings.